

C165

Integrated Physical Sciences

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A warehouse worker pushes a delivery cart that has a mass of 50 kg. The cart is painted red, and the warehouse floor is 200 meters long. The net force on the cart is 100 N. What is the cart's acceleration?

- A. 5,000 m/s²
- B. 2 m/s²
- C. 0.5 m/s²
- D. 100 m/s²

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A 1,200 kg car speeds up at 3 m/s². What net force acts on it?

- A. 1,200 N
- B. 400 N
- C. 3,600 N
- D. 1,203 N

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A student places a wooden block, cut from a tree in Oregon, into a tank that holds 50 liters of water. The block has a mass of 600 g and a volume of 800 cm³. Water's density is 1.0 g/cm³. What is the block's density, and does it float?

- A. 0.75 g/cm³; it floats.
- B. 1.33 g/cm³; it sinks.
- C. It floats only if the tank is large enough.
- D. It sinks, because 600 g is heavy for its size.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A liquid has a density of 0.8 g/cm^3 . What is the mass of 250 cm^3 of it?

- A.** 250 g
- B.** 312.5 g
- C.** 200 g
- D.** 0.0032 g

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A 2 kg rubber ball, bought at a sporting-goods store, is dropped from a height of 5 m on a calm afternoon. Ignore air resistance and use $g = 9.8 \text{ m/s}^2$. How much kinetic energy does the ball have just before it hits the ground?

- A.** 98 J
- B.** 10 J
- C.** 0 J
- D.** 49 J

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

In a restaurant kitchen that serves 300 meals a day, a cook melts butter, boils water for pasta, stirs sugar into iced tea until it disappears, and toasts bread until it turns brown. Which of these is a chemical change?

- A.** Dissolving the sugar, because anything that disappears has changed chemically.
- B.** Toasting the bread until it browns.
- C.** Boiling the water, because bubbles form.
- D.** Melting the butter.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

In a sealed flask, 10 g of baking soda reacts with 50 g of vinegar, and bubbles of gas form. The empty glass flask has a mass of 200 g and was made by a lab supplier in Ohio. What is the total mass of the flask's contents after the reaction?

- A. 60 g
- B. More than 60 g, because a new substance forms.
- C. 260 g
- D. Less than 60 g, because a gas is produced.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

The same reaction is run in an open beaker. The beaker and its contents weigh 5 g less afterward. What happened to the 5 g?

- A. It was destroyed in the reaction.
- B. It was converted into energy.
- C. It is a scale error, since a reading can never change in any container.
- D. It left the beaker as a gas.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A chef with 20 years of experience leaves a metal spoon in a pot of simmering soup. After a few minutes, the handle, which sticks out well above the soup, is too hot to touch. The soup is a family recipe. How did the heat reach the handle?

- A. Insulation.
- B. Radiation, since all heat moves as light.
- C. Convection, because the soup is a liquid.
- D. Conduction.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A student tests whether fertilizer affects tomato plant height. She gives groups of plants 0 g, 5 g or 10 g of fertilizer each week, keeps them all in the same greenhouse, built in 2010, and waters each plant the same amount. After four weeks she measures their heights. What is the independent variable?

- A.** The greenhouse.
- B.** The amount of fertilizer.
- C.** The amount of water.
- D.** Plant height.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A science teacher in Minnesota, whose class keeps a weather journal, points out that in early July Earth is near its farthest point from the Sun. Yet July is summer in Minnesota. What explains this?

- A.** Distance always sets the seasons, so the teacher must be wrong.
- B.** Earth is actually closest to the Sun in July.
- C.** The Northern Hemisphere is tilted toward the Sun, so sunlight is more direct and days are longer.
- D.** The Sun gives off more energy in the summer months.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

What season is it in Australia in December, and why?

- A.** Spring, since seasons are the same worldwide on any date.
- B.** Summer, because Earth is closest to the Sun in early January.
- C.** Winter, the same as in the Northern Hemisphere.
- D.** Summer, because the Southern Hemisphere is tilted toward the Sun.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A warehouse worker pushes a delivery cart that has a mass of 50 kg. ~~The cart is painted red, and the warehouse floor is 200 meters long.~~ The net force on the cart is 100 N. What is the cart's acceleration?

Target word: acceleration **Decides:** $a = F / m$.

A	Confusion set	Multiplies force by mass, which has no meaning here.
B	Correct	$100 \text{ N} / 50 \text{ kg} = 2 \text{ m/s}^2$.
C	Inverted operation	Divides mass by force.
D	Wrong stem number	That is the force, not the acceleration.

2. Answer C Concept 1 • transfer

A 1,200 kg car speeds up at 3 m/s^2 . What net force acts on it?

Target word: solve for force **Decides:** $F = m \times a$.

A	Wrong stem number	That is the mass in kilograms.
B	Inverted operation	Divides mass by acceleration.
C	Correct	$1,200 \times 3 = 3,600 \text{ N}$.
D	Confusion set	Adds mass and acceleration.

3. Answer A Concept 2

A student places a wooden block, ~~cut from a tree in Oregon,~~ into a tank ~~that holds 50 liters of water.~~ The block has a mass of 600 g and a volume of 800 cm^3 . Water's density is 1.0 g/cm^3 . What is the block's density, and does it float?

Target word: density and float/sink **Decides:** $600 / 800 = 0.75 \text{ g/cm}^3$, less than water.

A	Correct	Its density is less than water's.
B	Inverted operation	Divides volume by mass.
C	Camouflage	Tank size does not change whether an object floats.
D	Dropped element	Drops the volume; mass alone does not decide.

4. Answer C Concept 2 • transfer

A liquid has a density of 0.8 g/cm^3 . What is the mass of 250 cm^3 of it?

Target word: solve for mass **Decides:** $\text{Mass} = \text{density} \times \text{volume}$.

A	Wrong stem number	That is the volume, which equals mass only when density is 1.
B	Inverted operation	Divides volume by density.
C	Correct	$0.8 \times 250 = 200 \text{ g}$.
D	Confusion set	Divides density by volume.

5. Answer A Concept 3

A 2 kg rubber ball, ~~bought at a sporting goods store,~~ is dropped from a height of 5 m ~~on a calm afternoon.~~ Ignore air resistance and use $g = 9.8 \text{ m/s}^2$. How much kinetic energy does the ball have just before it hits the ground?

Target word: kinetic energy at the bottom **Decides:** All potential energy mgh becomes kinetic energy.

A	Correct	$2 \times 9.8 \times 5 = 98 \text{ J.}$
B	Dropped element	$2 \times 5 \text{ drops g.}$
C	Inverted operation	Kinetic energy is zero at the top, not the bottom.
D	Confusion set	Mixes in the $\frac{1}{2}$ from the kinetic energy formula.

6. Answer B Concept 4

In a restaurant kitchen ~~that serves 300 meals a day,~~ a cook melts butter, boils water for pasta, stirs sugar into iced tea until it disappears, and toasts bread until it turns brown. Which of these is a chemical change?

Target word: the chemical change **Decides:** Browning makes new substances.

A	Absolute language	Dissolved sugar is still sugar; evaporating the water gets it back.
B	Correct	Browning creates new compounds and cannot be undone.
C	Adjacent category	The bubbles are water vapor, still water; a change of state.
D	Confusion set	A change of state; it is still butter.

7. Answer A Concept 5

In a sealed flask, 10 g of baking soda reacts with 50 g of vinegar, and bubbles of gas form. The empty glass flask has a mass of 200 g ~~and was made by a lab supplier in Ohio.~~ What is the total mass of the flask's contents after the reaction?

Target word: mass of the contents after reaction **Decides:** The flask is sealed, so nothing escapes.

A	Correct	Mass is conserved in a closed system: $10 \text{ g} + 50 \text{ g.}$
B	Inverted operation	New substances do not add mass.
C	Camouflage	Adds the flask, which the question does not ask about.
D	Dropped element	Drops that the flask is sealed; the gas stays inside.

8. Answer D Concept 5 • transfer

The same reaction is run in an open beaker. The beaker and its contents weigh 5 g less afterward. What happened to the 5 g?

Target word: where the mass went **Decides:** The container is open.

A	Inverted operation	Chemical reactions do not destroy mass.
B	Confusion set	Measurable mass-to-energy conversion happens in nuclear reactions, not this one.
C	Absolute language	An open container can lose mass to escaping gas.
D	Correct	Mass is conserved; the escaping gas carried it away.

9. Answer D Concept 6

A chef ~~with 20 years of experience~~ leaves a metal spoon in a pot of simmering soup. After a few minutes, the handle, which sticks out well above the soup, is too hot to touch. ~~The soup is a family recipe.~~ How did the heat reach the handle?

Target word: heat transfer mode **Decides:** Heat moved through the solid metal.

A	Confusion set	Insulation slows heat transfer; it is not a mode of transfer.
B	Absolute language	Radiation is one of three modes, and not the main one here.
C	Adjacent category	Convection moves heat within the soup; the handle is above it.
D	Correct	Heat traveled through the spoon by direct contact between particles.

10. Answer B Concept 7

A student tests whether fertilizer affects tomato plant height. She gives groups of plants 0 g, 5 g or 10 g of fertilizer each week, keeps them all in the same greenhouse, ~~built in 2010~~, and waters each plant the same amount. After four weeks she measures their heights. What is the independent variable?

Target word: independent variable **Decides:** She deliberately changes the fertilizer amount.

A	Camouflage	The setting is held constant; it is not a variable being tested.
B	Correct	The factor the experimenter changes on purpose.
C	Adjacent category	Kept the same for all plants: a controlled variable.
D	Inverted operation	Height is measured: the dependent variable.

11. Answer C Concept 8

A science teacher in Minnesota, ~~whose class keeps a weather journal~~, points out that in early July Earth is near its farthest point from the Sun. Yet July is summer in Minnesota. What explains this?

Target word: cause of summer **Decides:** The Northern Hemisphere is tilted toward the Sun in July.

A	Absolute language	Distance has only a small effect; tilt dominates.
B	Inverted operation	Earth is closest in early January.
C	Correct	Axial tilt, not distance, causes seasons.
D	Confusion set	The Sun's output barely changes over the year.

12. Answer D Concept 8 • transfer

What season is it in Australia in December, and why?

Target word: Southern Hemisphere season **Decides:** In December the Southern Hemisphere is tilted toward the Sun.

A	Absolute language	Seasons are reversed between hemispheres.
B	Confusion set	Right season, wrong cause; if distance caused it, both hemispheres would have summer.
C	Inverted operation	The hemispheres have opposite seasons.
D	Correct	Tilt gives the Southern Hemisphere more direct sunlight in December.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	3	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Newton's second law	Net force = mass $\times$ acceleration ($F = ma$); acceleration = net force / mass.	Dividing the wrong way or multiplying when you should divide.	A mass and a net force, or a mass and an acceleration.
2	Density and floating	Density = mass / volume; an object floats in a fluid if its density is less than the fluid's.	Judging by mass or size alone.	A mass, a volume and a question about sinking or floating.
3	Conservation of energy	Without friction or air resistance, gravitational potential energy (mgh) at the top becomes kinetic energy at the bottom.	Leaving out g , or mixing in the $\frac{1}{2}$ from the kinetic energy formula.	An object dropped from a height, air resistance ignored.
4	Physical vs. chemical change	A chemical change makes new substances (signs: new color, gas, heat, light, odor that is not reversible); physical changes such as melting, boiling and dissolving do not.	Treating bubbles from boiling or a solid disappearing into water as chemical.	A list of everyday changes.
5	Conservation of mass	In a chemical reaction, total mass stays the same; mass seems to drop only if a gas escapes an open container.	Thinking gas production destroys mass in a sealed container.	Reactant masses and whether the container is sealed or open.
6	Heat transfer	Conduction moves heat through direct contact in solids; convection through moving fluids; radiation through electromagnetic waves.	Calling any heating that involves a liquid convection.	How the heat reaches the object: touch, flowing fluid, or across space.
7	Experimental variables	The independent variable is what the experimenter changes; the dependent variable is what is measured; controlled variables are kept the same.	Swapping the independent and dependent variables.	An experiment with one factor changed on purpose and one thing measured.
8	Cause of the seasons	Seasons come from Earth's axial tilt, which changes how directly sunlight strikes each hemisphere and how long days are, not from distance to the Sun.	Believing Earth is closer to the Sun in summer.	A season question that mentions Earth's distance from the Sun.

Course scope (public description): This course provides students with an overview of the basic principles and unifying ideas of the physical sciences: physics, chemistry, and earth sciences. Course materials focus on scientific reasoning and practical, everyday applications of physical science concepts to help students integrate conceptual knowledge with practical skills.